

Feature	Router	Switch
Main use	Connects different networks (such as a home network to the internet).	Connects multiple devices within the same local network (LAN).
Typical location	Found at the gateway between a local network and the internet, often provided by an ISP or used in homes and offices.	Located within a home, office, or data center LAN to connect computers, printers, servers, and other devices.
How they handle data	Forwards data packets between different networks using IP addresses and routing tables.	Forwards data frames within a LAN using MAC addresses to send data directly to the intended device.
OSI layer	Primarily operates at Layer 3 (Network Layer) .	Primarily operates at Layer 2 (Data Link Layer) , though some switches also support Layer 3 features.
Address used	Uses IP addresses to determine the best path for packets.	Uses MAC addresses to identify and forward data to devices on the local network.
Traffic management	Chooses the best route between networks and can perform functions like NAT and firewalling.	Reduces unnecessary traffic by sending data only to the destination port instead of broadcasting to all devices.